

Quiz details

Quiz instructions:

This quiz will focus on the material from Week 9 in the lab workbook. To answer the following questions, you will need to use the [dive_data.csv](https://canvas.stir.ac.uk/courses/19303/files/4997287/download?download_frd=1)  (https://canvas.stir.ac.uk/courses/19303/files/4997287/download?download_frd=1) dataset and the [Angola_soils.csv](https://canvas.stir.ac.uk/courses/19303/files/4997326/download?download_frd=1)  (https://canvas.stir.ac.uk/courses/19303/files/4997326/download?download_frd=1) datasets.

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Diving data

Assume that you have been out in the field to observe the diving behaviour of 3 species of pinnipeds, which you have recorded. Specifically, you recorded diving behaviours at different times of the day (dive_time; day and night) of grey seal, California sea lions, and Southern sea lions. Use the [dive_data](https://canvas.stir.ac.uk/courses/19303/files/4997287?wrap=1) (<https://canvas.stir.ac.uk/courses/19303/files/4997287?wrap=1>)  (https://canvas.stir.ac.uk/courses/19303/files/4997287/download?download_frd=1) file provided to answer questions 1-4.

Question 1 1 pts

If pinnipeds equally likely to dive during the day and during the night, then what should be the expected count of day dives for this dataset? Write your answer as an integer.

61 (with margin: 0)

Correct answer comments

There are 122 observations in this dataset, and dive time can be either day or night (2 options). Consequently, if day and night dives are equally likely, then we would expect the count of each to be $122/2 = 61$.

Wrong answer comments

There are 122 observations in this dataset, and dive time can be either day or night (2 options). Consequently, if day and night dives are equally likely, then we would expect the count of each to be $122/2 = 61$.

Question 2 1 pts

Test the null hypothesis that there is no significant difference between the expected and observed dive time counts, assuming that the expected counts of day and night dives are the same. From the Chi-square goodness of fit test in Jamovi, what should you conclude?

- ☐ Reject the null hypothesis because the Chi-square test statistic is greater than 0.05
- ☐ Do not reject the null hypothesis because the Chi-square test statistic is greater than 0.05
- ☐ Reject the null hypothesis because the p-value is greater than 0.05
- ☐ Do not reject the null hypothesis because the p-value is greater than 0.05
- ☐ Reject the null hypothesis because the Chi-square test statistic is greater than the p-value
- ☐ Do not reject the null hypothesis because the Chi-square test statistic is greater than the p-value

Correct answer comments

To answer this question, you need to navigate to the analysis tab in Jamovi and selection 'Frequencies', then 'N Outcomes: Chi-square goodness of fit test' in the pulldown menu. In the panel that opens up, put 'Dive_time' into the variable box. The Chi-square goodness of fit table will appear in the right panel. The Chi-square statistic is 0.52459, and the p-value is 0.46889. As always, we reject the null hypothesis if $P < 0.05$.

Wrong answer comments

To answer this question, you need to navigate to the analysis tab in Jamovi and selection 'Frequencies', then 'N Outcomes: Chi-square goodness of fit test' in the pulldown menu. In the panel that opens up, put 'Dive_time' into the variable box. The Chi-square goodness of fit table will appear in the right panel. The Chi-square statistic is 0.52459, and the p-value is 0.46889. As always, we reject the null hypothesis if $P < 0.05$.

Question 3 1 pts

Use a Chi-square goodness of fit test to see if observations are equally frequent for all 3 species of pinnipeds. What is the p-value of this test? Report your answer to 2 or more decimal places.

0.0119 (with margin: 0)

Correct answer comments

To answer this question, you need to navigate to the analysis tab in Jamovi and selection 'Frequencies', then 'N Outcomes: Chi-square goodness of fit test' in the pulldown menu. In the panel that opens up, put 'Species' into the variable box. The Chi-square goodness of fit table will appear in the right panel along with the p-value. The Chi-square statistic is 8.86885, and the p-value is 0.01186. Because $P < 0.05$, we should reject the null hypothesis that expected dive frequencies (40.666 for all species) are the same as observed dive frequencies.

Wrong answer comments

To answer this question, you need to navigate to the analysis tab in Jamovi and selection 'Frequencies', then 'N Outcomes: Chi-square goodness of fit test' in the pulldown menu. In the panel that opens up, put 'Species' into the variable box. The Chi-square goodness of fit table will appear in the right panel along with the p-value. The Chi-square statistic is 8.86885, and the p-value is 0.01186. Because $P < 0.05$, we should reject the null hypothesis that expected dive frequencies (40.666 for all species) are the same as observed dive frequencies.

Question 4 1 pts

Now, test the null hypothesis that there is no association between species and dive time. What is the Chi-square test statistic? Report your answer to 2 or more decimal

places.

2.3348 (with margin: 0.02)

Correct answer comments

To answer this question, you need to navigate to the analysis tab in Jamovi and selection 'Frequencies', then 'Independent samples: Chi-square test of association' in the pulldown menu. In the panel that opens up, put 'Species' into the rows box and Dive_time in the columns box (the order doesn't matter, reversing these is fine). The contingency tables will appear in the right panel. The Chi-square statistic is 2.33476, and the p-value is 0.31118. Because $P > 0.05$, we should not reject the null hypothesis that there is no association between species and dive time.

Wrong answer comments

To answer this question, you need to navigate to the analysis tab in Jamovi and selection 'Frequencies', then 'Independent samples: Chi-square test of association' in the pulldown menu. In the panel that opens up, put 'Species' into the rows box and Dive_time in the columns box (the order doesn't matter, reversing these is fine). The contingency tables will appear in the right panel. The Chi-square statistic is 2.33476, and the p-value is 0.31118. Because $P > 0.05$, we should not reject the null hypothesis that there is no association between species and dive time.

Question 5 1 pts

Again, test the null hypothesis that there is no association between species and dive time. What is the most appropriate conclusion from this test?

- ☐ Reject the null hypothesis because the Chi-square value is less than 0.05
- ☐ Do not reject the null hypothesis because the Chi-square value is greater than 0.05
- ☐ Reject the null hypothesis because the p-value is less than 0.05
- ☐ Do not reject the null hypothesis because the p-value is less than 0.05
- ☐ Reject the null hypothesis because the p-value is greater than 0.05
- ☐ Do not reject the null hypothesis because the p-value is greater than 0.05

Correct answer comments

To answer this question, you need to navigate to the analysis tab in Jamovi and selection 'Frequencies', then 'Independent samples: Chi-square test of association' in the pulldown menu. In the panel that opens up, put 'Species' into the rows box and Dive_time in the columns box (the order doesn't matter, reversing these is fine). The contingency tables will appear in the right panel. The Chi-square statistic is 2.33476, and the p-value is 0.31118. Because $P > 0.05$, we should not reject the null hypothesis that there is no association between species and dive time.

Wrong answer comments

To answer this question, you need to navigate to the analysis tab in Jamovi and selection 'Frequencies', then 'Independent samples: Chi-square test of association' in the pulldown menu. In the panel that opens up, put 'Species' into the rows box and Dive_time in the columns box (the order doesn't matter, reversing these is fine). The contingency tables will appear in the right panel. The Chi-square statistic is 2.33476, and the p-value is 0.31118. Because $P > 0.05$, we should not reject the null hypothesis that there is no association between species and dive time.

Colleague observations

Suppose that a colleague has observed pinnipeds in a completely different location, but they do not know how to create a tidy dataset or run a Chi-square goodness of fit test. They just give you there observations for each species:

California sea lion: 15

Grey seal: 6

Southern sea lion: 8

Based on these counts, run a goodness of fit test in Jamovi (hint: see [the last exercise \(https://bradduthie.github.io/SCIU4T4/Chapter_30.html#untidy-goodness-of-fit\)](https://bradduthie.github.io/SCIU4T4/Chapter_30.html#untidy-goodness-of-fit) in the Chapter 30 lab practical for an untidy way of doing the test).

Question 6 1 pts

Refer to the 'Colleague observations' above to answer this question. What is the Chi-square test statistic from the goodness of fit test? Write your answer to 2 or more decimal places.

4.6207 (with margin: 0.02)

Correct answer comments

To answer this question, you first needed to create a new dataset in Jamovi with 2 columns, one for 'Species' and the other for 'Counts'.

Counts of different species of pinnipeds observed.

Species	Counts
California sea lion	15
Grey seal	6
Southern sea lion	8

With these data input into Jamovi, you then needed to navigate to the analysis tab in and selection 'Frequencies', then 'N Outcomes: Chi-square goodness of fit test' in the pulldown menu. In the panel that opens up, put 'Species' into the variable box, then 'Counts' in the 'Counts (optional)' box. The Chi-square goodness of fit table will appear in the right panel along with the p-value. The Chi-square statistic is 4.62069, and the p-value is 0.09923. Because $P > 0.05$, we should not reject the null hypothesis that expected dive frequencies are the same as observed dive frequencies. In plain language, there is no evidence that dive frequencies differ among the 3 species.

Wrong answer comments

To answer this question, you first needed to create a new dataset in Jamovi with 2 columns, one for 'Species' and the other for 'Counts'.

Counts of different species of pinnipeds observed.

Species	Counts
California sea lion	15
Grey seal	6
Southern sea lion	8

With these data input into Jamovi, you then needed to navigate to the analysis tab in and selection 'Frequencies', then 'N Outcomes: Chi-square goodness of fit test' in the pulldown menu. In the panel that opens up, put 'Species' into the variable box, then 'Counts' in the 'Counts (optional)' box. The Chi-square goodness of fit table will appear in the right panel along with the p-value. The Chi-square statistic is 4.62069, and the p-value is 0.09923. Because $P > 0.05$, we should not reject the null hypothesis that expected dive frequencies are the same as observed dive frequencies. In plain language, there is no evidence that dive frequencies differ among the 3 species.

Question 7 1 pts

Refer to the 'Colleague observations' above to answer this question. Based on the Chi-square goodness of fit test, what do you tell your colleague?

- ☐ There is no evidence that there is a difference in dive frequency among the 3 species
- ☐ There is strong evidence that there is a difference in dive frequency among the 3 species
- ☐ California sea lions clearly dive more frequently than the other two species
- ☐ A Chi-square goodness of fit test is not possible with these data

Correct answer comments

To answer this question, you first needed to create a new dataset in Jamovi with 2 columns, one for 'Species' and the other for 'Counts'.

Counts of different species of pinnipeds observed.

Species	Counts
California sea lion	15
Grey seal	6
Southern sea lion	8

With these data input into Jamovi, you then needed to navigate to the analysis tab in and selection 'Frequencies', then 'N Outcomes: Chi-square goodness of fit test' in the pulldown menu. In the panel that opens up, put 'Species' into the variable box, then 'Counts' in the 'Counts (optional)' box. The Chi-square goodness of fit table will appear in the right panel along with the p-value. The Chi-square statistic is 4.62069, and the p-value is 0.09923. Because $P > 0.05$, we should not reject the null hypothesis that expected dive frequencies are the same as observed dive frequencies. In plain language, there is no evidence that dive frequencies differ among the 3 species.

Wrong answer comments

To answer this question, you first needed to create a new dataset in Jamovi with 2 columns, one for 'Species' and the other for 'Counts'.

Counts of different species of pinnipeds observed.

Species	Counts
California sea lion	15
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With these data input into Jamovi, you then needed to navigate to the analysis tab in and selection 'Frequencies', then 'N Outcomes: Chi-square goodness of fit test' in the pulldown menu. In the panel that opens up, put 'Species' into the variable box, then 'Counts' in the 'Counts (optional)' box. The Chi-square goodness of fit table will appear in the right panel along with the p-value. The Chi-square statistic is 4.62069, and the p-value is 0.09923. Because $P > 0.05$, we should not reject the null hypothesis that expected dive frequencies are the same as observed dive frequencies. In plain language, there is no evidence that dive frequencies differ among the 3 species.

Angola soils

For the next 3 questions, use the [Angola soils \(https://canvas.stir.ac.uk/courses/19303/files/4997326?wrap=1\)](https://canvas.stir.ac.uk/courses/19303/files/4997326?wrap=1)  (https://canvas.stir.ac.uk/courses/19303/files/4997326/download?download_frd=1) dataset from the Week 7 practical ([Chapter 27](#)  (https://bradduthie.github.io/SCIU4T4/Chapter_27.html) in the workbook).

Question 8 1 pts

Suppose that you are interested in testing whether or not different pairs of nutrient concentrations (Nitrogen, Potassium, and Phosphorus) are significantly correlated. For which combination of variables would a Pearson's product moment correlation test be appropriate?

- ☐ Nitrogen and Potassium
- ☐ Nitrogen and Phosphorus
- ☐ Potassium and Phosphorus
- ☐ Any combination of nutrient variables would be appropriate
- ☐ No combination of nutrient variables would be appropriate

Correct answer comments

To answer this question, we need to run a Shapiro-Wilk test of normality on each of the 3 nutrient variables. In Jamovi, go to Exploration, then choose Descriptives from the pulldown menu. Place all 3 nutrient variables (Nitrogen, Potassium, and Phosphorus) into the 'Variables' box, then select 'Shapiro-Wilk' test from the Statistics pulldown menu below. A table of Descriptives will appear and reveal that for Nitrogen and Phosphorus, $P > 0.05$, so a Pearson's product moment correlation is appropriate for these two variables. In contrast, the p-value for Potassium is much less than 0.05, meaning that it is not normally distributed. Consequently, a Spearman rank correlation coefficient would be more appropriate.

Wrong answer comments

To answer this question, we need to run a Shapiro-Wilk test of normality on each of the 3 nutrient variables. In Jamovi, go to Exploration, then choose Descriptives from the pulldown menu. Place all

3 nutrient variables (Nitrogen, Potassium, and Phosphorus) into the 'Variables' box, then select 'Shapiro-Wilk' test from the Statistics pulldown menu below. A table of Descriptives will appear and reveal that for Nitrogen and Phosphorus, $P > 0.05$, so a Pearson's product moment correlation is appropriate for these two variables. In contrast, the p-value for Potassium is much less than 0.05, meaning that it is not normally distributed. Consequently, a Spearman rank correlation coefficient would be more appropriate.

Question 9 1 pts

Run the most appropriate test to test the null hypothesis that Nitrogen and Potassium concentration are not correlated. What is the p-value of this test? Report your answer to 2 or more decimal places.

0.9402 (with margin: 0.02)

Correct answer comments

Because Potassium is not normally distributed, a Spearman rank correlation coefficient test is needed. To run this test, navigate to the Analyses tab in Jamovi and choose 'Regression' from the tool bar, then select 'Correlation Matrix' from the pulldown menu. Place Nitrogen and Potassium into the box on the right. Make sure that only 'Spearman' is selected. From the output table, the p-value will be 0.94024.

Wrong answer comments

Because Potassium is not normally distributed, a Spearman rank correlation coefficient test is needed. To run this test, navigate to the Analyses tab in Jamovi and choose 'Regression' from the tool bar, then select 'Correlation Matrix' from the pulldown menu. Place Nitrogen and Potassium into the box on the right. Make sure that only 'Spearman' is selected. From the output table, the p-value will be 0.94024.

Question 10 1 pts

Run the most appropriate test to test the null hypothesis that Nitrogen and Phosphorus concentration are not correlated. What should you conclude from this test and why?

- ☐ There is no evidence that Nitrogen and Phosphorus concentrations are correlated
- ☐ There is evidence that Nitrogen and Phosphorus concentrations are correlated
- ☐ There is evidence that Nitrogen and Phosphorus are not normally distributed
- ☐ There is evidence that Nitrogen and Phosphorus are negatively correlated
- ☐ None of the above choices are correct

Correct answer comments

Because Nitrogen and Phosphorus are both normally distributed, a Pearson product moment correlation is most appropriate to use. To run this test, navigate to the Analyses tab in Jamovi and choose 'Regression' from the tool bar, then select 'Correlation Matrix' from the pulldown menu. Place Nitrogen and Phosphorus into the box on the right. Make sure that only 'Pearson' is selected. From the output table, the correlation coefficient (r) is 0.01253, and the p-value is 0.93188. We therefore should not reject the null hypothesis that the data are not correlated.

Wrong answer comments

Because Nitrogen and Phosphorus are both normally distributed, a Pearson product moment correlation is most appropriate to use. To run this test, navigate to the Analyses tab in Jamovi and choose 'Regression' from the tool bar, then select 'Correlation Matrix' from the pulldown menu. Place Nitrogen and Phosphorus into the box on the right. Make sure that only 'Pearson' is selected. From the output table, the correlation coefficient (r) is 0.01253, and the p-value is 0.93188. We therefore should not reject the null hypothesis that the data are not correlated.